INSTITUT PENYELIDIKAN PERHUTANAN MALAYSIA
Forest Research Institute Malaysia (FRIM)
52109 Kepong, Selangor
Tel: 603-6279 7000 Fax: 603 6273 1314
Portal Rasmi: www.frim.gov.my

Your reference:
Our reference: FRIM (S).600-5/6/5 Klt. 12(39)
Date : 4 September 2020

CONFIDENTIAL

Reimagine ME Sdn. Bhd,
21-1 Jalan Putera Mahkota
7/6 B, Seksyen 7, Putera Height
47650 Subang Jaya, Selangor

(Attn. Mr. Holy Lim)

Sir,

EFB STRAND BOARD (OPLY)

Referring to the above, enclosed is report of EFB Strand Board (OPLY) sample tested for bending, internal bond, surface soundness, screw withdrawal, thickness swelling and water absorption.

Thank you.

Yours sincerely,

(.....)
DR. SUFFIAN MISRAN
for Director General
FRIM

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MAKMAL PENGUJIAN MEKANIKAL & FIZIKAL (MPTL)
INSTITUT PENYELIDIKAN PERHUTAHAN MALAYSIA (FRIM)
52109 KEPONG, SELANGOR DARUL EHSAN
Tel: 603-6279 7288/7310 Fax: 603 62804620/4629
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MS ISO/IEC 17025
TESTING
SAMM NO. 205

Our reference: FRIM (S).600-5/6/5 Klt. 12(39)

Date : 4 September 2020

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TEST REPORT

Job No.: 23/ 20	
This report consists of 7 pages	Page 1 of 7

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CUSTOMER

Name and Address: Reimagine ME Sdn. Bhd.
21-1 Jalan Putera Mahkota
7/6 B, Seksyen 7, Putera Height
47650 Subang Jaya, Selangor

SAMPLE DESCRIPTION

WCTL serial number: 1240620

Type of material & identity: EFB Strand Board (OPLY)
Thickness – 12mm

Date received: 27 July 2020

Note:
Test specimens were prepared by customer. The test specimens will be disposed after testing has been carried out.

TEST METHOD

Tested for and standard method used:

1. Bending (BS EN 310: 1993)
2. Internal bond (BS EN 319: 1993)*
3. Surface soundness (BS EN 311: 2002)*
4. Screw withdrawal (BS EN 320: 2011)*
5. Thickness swelling (BS EN 317:1993)*
6. Water absorption (In-house method)*

Date of test: 28 July 2020 – 6 August 2020

*...Not SAMM Accredited

Prepared by

Nordin Puteh
Assistant Research Officer

Approved by

Dr. Suffian Misran
Research Officer

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TEST RESULTS

Table 1: Bending (BS EN 310: 1993) – EFB Strand Board (OPLY)

Specimen	Modulus of Rupture (N/mm^2)	Modulus of Elasticity (N/mm^2)
1	28.5	1740
2	23.8	1570
3	38.2	2220
4	34.5	2220
5	36.2	2110
6	32.2	1860
7	25.4	1600
8	31.6	1720
9	37.4	2060
10	34.1	1950
Average	**32.2**	**1910**
Std. Deviation	**4.9**	**244**

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TEST RESULTS

Table 2: Internal Bond (BS EN 319: 1993) – EFB Strand Board (OPLY)

Specimen	Internal Bond (N/mm^2)
1	1.90
2	1.73
3	1.26
4	0.70
5	0.84
6	0.60
7	1.85
8	0.56
9	0.60
10	1.54
Average	**1.16**
Std. Deviation	**0.56**

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TEST RESULTS

Table 3: Surface Soundness (BS EN 311: 2002) – EFB Strand Board (OPLY)

Specimen	Surface Soundness (N/mm^2)
1	2.92
2	1.71
3	3.00
4	2.80
5	1.42
6	2.46
7	2.48
8	3.21
9	2.57
10	2.08
Average	**2.46**
Std. Deviation	**0.58**

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TEST RESULTS

Table 4: Screw Withdrawal (BS EN 320: 2011) – EFB Strand Board (OPLY)

Specimen	Screw Withdrawal (Face) (N/mm)
1	138
2	148
3	140
4	105
5	152
6	187
7	187
8	181
9	252
10	248
Average	**174**
Std. Deviation	**48**

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TEST RESULTS

Table 5: Thickness Swelling (BS EN 317: 1993) – EFB Strand Board (OPLY)

Specimen	Thickness Swelling (%)
1	7.2
2	6.2
3	5.9
4	6.8
5	6.1
6	6.3
7	6.5
8	6.9
9	6.1
10	6.3
Average	**6.4**
Std. Deviation	**0.4**

Note: Specimens were soaked in 20°C water for 24 hours

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TEST RESULTS

Table 6: Water Absorption (In-house method) – EFB Strand Board (OPLY)

Specimen	Water Absorption (%)
1	58.6
2	39.5
3	31.0
4	53.8
5	33.7
6	33.2
7	58.2
8	61.1
9	30.2
10	28.8
Average	**42.8**
Std. Deviation	**13.4**

Note: 50mm x 50mm specimens were soaked in 20°C water for 24 hours, followed by weight increment determination

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